

Part1

1)Predictor Variable1:Race(white/black/other)

Predictor Variable2:R is how tall

Predictor Variable3:Day of poor mental health past 30 days

Predictor Variable4:High year of school completed

Predictor Variable5:R can speak language other than English/Spanish(yes/no)

dependent Variable: Family income in constant

H0: The combination of predictor variable are not linearly related to the dependent variable

H1: The combination of predictor variable are linearly related to the dependent variable

Variable Creation

	Label
Otherlanguage_1	OTHLANG=YES
Otherlanguage_2	OTHLANG=NO
Otherlanguage_3	OTHLANG=DK

Case Processing Summary

	Valid		Cases Missing		Total	
	N	Percent	N	Percent	N	Percent
Can R speak language other than English/Spanish * OTHLANG=YES	2346	99.9%	2	0.1%	2348	100.0%

Can R speak language other than English/Spanish * OTHLANG=YES Crosstabulation

Count

		OTHLANG=YES		Total
		.00	1.00	
Can R speak language other than English/Spanish	YES	0	723	723
	NO	1623	0	1623
Total		1623	723	2346

Model Summary^b

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.428 ^a	.184	.180	28390.182

a. Predictors: (Constant), OTHLANG=YES, R is how tall, Days of poor mental health past 30 days, Highest year of school completed, RACE=BLACK, RACE=OTHER

b. Dependent Variable: Family income in constant \$

The Adjusted R Square is 0.18, that means the 18% variance in the dependent variable family income.

2)

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	2.381E+11	6	39683300438	49.235	<.001 ^b
	Residual	1.059E+12	1314	806002451.92		
	Total	1.297E+12	1320			

a. Dependent Variable: Family income in constant \$

b. Predictors: (Constant), OTHLANG=YES, R is how tall, Days of poor mental health past 30 days, Highest year of school completed, RACE=BLACK, RACE=OTHER

The ANOVA table showed that the regression model was statistically significant ($F=49.235$, $P<0.001$). This indicates that the predictor variables, including race, education, height, mental health days, and speaking another language, collectively explained a significant amount of variation in family income. The p-value of less than 0.001 means that there is less than a 0.1% probability of obtaining a relationship this strong if the null hypothesis were true. Therefore, the null hypothesis that there is no linear relationship between the predictors and family income was rejected. This suggests that the regression model provides a significant prediction of family income.

3)

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	-94972.850	13499.217		-7.035	<.001		
	RACE=BLACK	-10441.950	2140.289	-.124	-4.879	<.001	.960	1.042
	RACE=OTHER	-3737.394	2663.380	-.038	-1.403	.161	.845	1.184
	R is how tall	1244.485	192.218	.164	6.474	<.001	.969	1.032
	Days of poor mental health past 30 days	-391.220	113.803	-.086	-3.438	<.001	.985	1.015
	Highest year of school completed	3727.024	283.971	.333	13.125	<.001	.963	1.039
	OTHLANG=YES	1271.875	1795.485	.019	.708	.479	.894	1.119

a. Dependent Variable: Family income in constant \$

The significance values (p-values) show whether each predictor variable has a statistically significant relationship with family income.

(1) Variables with $p < 0.05$ are considered statistically significant:

RACE=BLACK: $p < 0.001 \rightarrow$ significant negative relationship with income

R is how tall: $p < 0.001 \rightarrow$ significant positive relationship

Days of poor mental health past 30 days: $p < 0.001 \rightarrow$ significant negative relationship

Highest year of school completed: $p < 0.001 \rightarrow$ strongest significant positive relationship

(2) Variables that are not statistically significant ($p > 0.05$):

RACE=OTHER: $p = 0.161 \rightarrow$ not significant

OTHLANG=YES: $p = 0.479 \rightarrow$ not significant

Therefore, we reject the null hypothesis for variables with $p < 0.05$ and conclude they are significantly associated with family income. For variables with $p > 0.05$, we fail to reject the null hypothesis.

4) The largest absolute standardized coefficients (Beta) indicate the most important predictors in the model:

Highest year of school completed: Beta = 0.333

$\rightarrow$ strongest positive predictor of family income

R is how tall: Beta = 0.164

$\rightarrow$ second strongest positive predictor

RACE=BLACK: Beta = -0.124

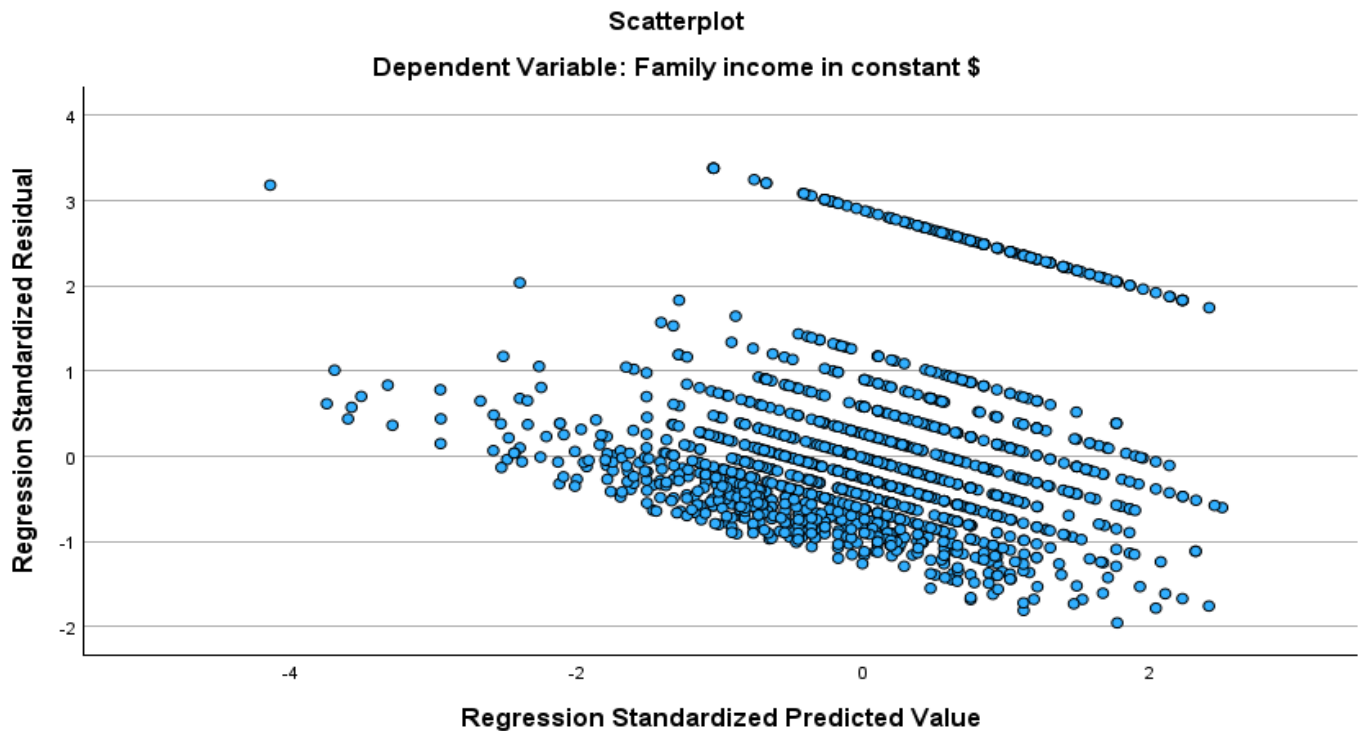
$\rightarrow$ strongest negative predictor

Because these variables have the largest absolute Beta values, they have the greatest impact on family income relative to the other predictors in the model.

5) The unstandardized coefficient for highest year of school completed is 3727.024. This means that for each additional year of education completed, predicted family income increases by about \$3,727, holding all other variables constant.

6) all the VIF statistics for every variable were close to 1, and so we don't have a problem with collinearity.

7)



The scatterplot shows some pattern in the residuals rather than a completely random and even spread.

In this plot, the residuals appear somewhat clustered and form diagonal patterns, with greater spread at some predicted values than others. There are also several possible outliers with large residuals above 3 and below -2.

Therefore, the assumption of equal variances may be somewhat violated, because the residuals are not perfectly uniformly distributed across the predicted values. However, most points are still concentrated around zero residuals.

Part2

The purpose of this multiple linear regression exercise was to examine whether several predictor variables were significantly associated with family income and to determine which variables were the strongest predictors in the model. The dependent variable was family income, while the independent variables included race, education, height, days of poor mental health, and whether the respondent could speak another language besides English or Spanish. The ANOVA results showed that the overall regression model was statistically significant, $F = 49.235$, $p < 0.001$, meaning the combination of predictor variables explained a significant amount of variation in family income. The model summary indicated that the predictors explained approximately 18.0% of the variance in family income based on the adjusted R-squared value (Adjusted $R^2 = 0.180$). Several predictors were statistically significant, including highest year of school completed, height, days of poor mental health, and race=Black. The strongest standardized coefficient was highest year of school completed (Beta = 0.333), indicating it was the most important predictor in the model. The residual scatterplot showed that the residuals were somewhat unevenly distributed across predicted values, suggesting that the equal variance assumption may have been partially violated. Overall, the regression model demonstrated meaningful but moderate predictive ability for explaining differences in family income.